Math 241, F12 **Name:** Answer

Quiz 1 **SSN:** xxx-xx-

<u>Instructions:</u> There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz and exam.

Problem 1. (6 points) Find $3\mathbf{a} + 2\mathbf{b}$, $|\mathbf{a}|$ and $|\mathbf{a} - \mathbf{b}|$ with the vectors $\mathbf{a} = \mathrm{i} + 2\mathrm{j} - 3\mathbf{k}$ and $\mathbf{b} = -2\mathrm{i} - 2\mathrm{j} + 9\mathbf{k}$.

$$3a+2b = 3(i+2j-3k)+2(-2i-2j+9k)$$
$$= 3i+6j-9k-4i-4j+18k$$
$$= -i+2j+9k$$

$$|a| = \sqrt{1^2+2^2+(-3)^2} = \sqrt{14}$$

$$|a-b| = |(i+2j-3k)-(-2i-2j+9k)|$$
$$= |3i+4j-12k|$$
$$= \sqrt{3^2+4^2+(-12)^2}$$
$$= \sqrt{169}$$
$$= 13$$

Problem 2. (4 points) Find a unit vector that has the same direction as $< -2, 4, 4 >$.

let $a = <-2,4,4>$

$$|a| = \sqrt{(-2)^2+4^2+4^2} = \sqrt{36} = 6$$

thus the unit vector having the same direction as a is given by

$$u = \frac{a}{|a|} = \frac{<-2,4,4>}{6} = <-\frac{1}{3}, \frac{2}{3}, \frac{2}{3}>$$